

Week 4 Assignment Submission PDF Content

Title:

Week 4 – RAG Project Setup Completion

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1. What I Set Up This Week

- Created and activated a Python virtual environment (venv)
 - Created requirements.txt with the required dependencies (fastapi, uvicorn, python-dotenv, google-generativeai)
 - Installed all dependencies using `pip install -r requirements.txt`
 - Created .env file to securely store my Gemini API key
 - Updated .gitignore to protect sensitive files (.env, venv/, __pycache__/)
 - Created rag_app.py as the main application file
 - Successfully ran the FastAPI server using uvicorn
 - Tested the /health endpoint at <http://127.0.0.1:8000/health> - successful
 - Committed all changes and pushed to GitHub-successful
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2. Purpose of rag_app.py

This file is the main backbone of the RAG application.

At this stage (Week 4), it:

- Loads the Gemini API key from the .env file using python-dotenv
 - Creates a FastAPI web application
 - Provides a /health endpoint to confirm the app is running correctly
 - Checks whether the API key was successfully loaded
 - Prepares the foundation for adding RAG logic in future weeks
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3. Proof of Completion

- The app runs locally without errors
- The health check page returns: `{"status":"ok","message":"RAG app is running!"}`
- Gemini API key is properly loaded from .env
- All dependencies are installed

- Code is committed and pushed to GitHub (see commit message below)

GitHub Commit Message:

"Week 4: Basic project setup with FastAPI, .env, and health endpoint"

4. Questions / Uncertainties

- Still learning how FastAPI and virtual environments work (this is all new to me)
- Need more practice understanding the connection between .env, load_dotenv(), and using the API key in code
- Ready to start adding actual Gemini calls and RAG functionality in Week 5